JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B2, PAGES 1949–1962, FEBRUARY 10, 1991

Paleomagnetism of the Silurian Volcanism at Almaden, Southern Spain

HERVÉ PERROUD, FRANÇOISE CALZA AND DRISS KHATTACH

Laboratoire de Géophysique Interne, Campus de Beaulieu, Université de Rennes 1, Rennes, France

The Silurian volcanism of the Almaden mining district, southern Spain, has been the object of a very detailed paleomagnetic study. The 175 cores (25 sites) sampled all along the eastern termination of the Almaden syncline were subjected to magnetic susceptibility measurements, thermomagnetic cycles, progressive alternating field and thermal demagnetizations as well as microscopic observations. The magnetic mineralogy of these volcanics flows is dominated by magnetite in at least two very distinct forms, including small cubic distributed grains and large elongated crystals. The observed anisotropy of magnetic susceptibility varies according to the relative abundance of the two magnetite phases, showing a complete inversion of the susceptibility tensors. This anisotropy has been related to the emplacement of the flows on a gentle slope dipping ~ 10° toward the southwest, implying that their volcanic source was located to the northeast. The paleomagnetic analysis has revealed two characteristic components of magnetization: the first (direction D=109°, I=+11°, α_{95}=9°, pole position at 11°S, 69°E, dp=5°, dm=9°) yields a negative fold test and is interpreted as a Late Paleozoic partial overprint; its declination is 46° away of the expected Late Paleozoic field direction, which confirms that rotations are still occurring after the main folding phases of the Variscan orogeny; the second (direction D=62°, I=-36°, α_{95}=14°, pole position at 8°N, 118°E, dp=9°, dm=13°) is prefolding as evident from the significant clustering when applying appropriate tectonic corrections, and we interprete it as the original magnetization of Silurian age. The compilation of all Paleozoic paleomagnetic results from Spain reveals that the Iberian Meseta has been moving along with Gondwana at high drift rates throughout Ordovician and Silurian times.

INTRODUCTION

Paleomagnetic data from the western Europe Variscan basement have given significant insights into the paleogeographic arrangement of continental landmasses in Ordovician and Late Paleozoic times [*Briden et al.*, 1984; *Perroud et al.*, 1984]. They reveal the evolution of the "Armorican" terranes [*Van der Voo*, 1979] initially situated at the margin of the West African craton, and then accreted to Eurasia. However, major questions remain unanswered: What was the timing of the docking? Was "Armorica" a separate plate, and if so what was its extent? New data from intermediate periods are necessary to resolve such questions. In this paper we report new paleomagnetic results from the Silurian volcanism of the Almaden mining district, situated in southern Spain. As no fully reliable data are available for the Silurian period from southern Europe, it was extremely important to conduct a very detailed paleomagnetic analysis of this volcanic sequence. Preliminary results, from two sites, were reported in 1967 by *Van der Voo* [1967] showing a complex magnetization but also evidence for pretectonic components. That encouraged us to start a new and more comprehensive investigation.

GEOLOGICAL CONTEXT AND SAMPLING

The Silurian volcanics outcrop along the border of the Almaden syncline, in the southern part of the Iberian Meseta (Central Iberian Zone). This area is characterized by a set of large Paleozoic synclines separated by Precambrian anticlines, and it can be closely correlated across the Bay of Biscaye with the Central Armorican Zone of the Armorican Massif [*Paris and Robardet*, 1977]. The Almaden syncline has long been the object of geological studies since the discovery of large ore deposits, still in exploitation today. The Silurian strata here comprise a 700m-thick sequence, half of it volcanic. The age of the volcanism is inferred from graptolites found in intercalated sediments, belonging to the Llandovery and Wenlock stages [*Saupé*, 1971]. The rocks were described as being mainly spilitized basaltic flows, or pyroclastic tuffs, errupted from a submarine volcano.

The Almaden syncline is a Variscan structure, oriented WNW-ESE. Its central part is more or less cylindrical, but the eastern termination is more complex, with conical folds and steeply plunging fold axes [*Verges*, 1983]. A mean fold axis has been determined by *Verges* [1983], with a 60° plunge in direction 304° (NW). The folding was attributed to the Sudetian phase of the Variscan orogeny (325 Ma), before the intrusion of the Fontanosas granite, 20 km east of Almaden, dated at 302 Ma [*Leutwein et al.*, 1970].

Sampling was conducted along the eastern border of the syncline (Figure 1). One hundred seventy-five cores from 25 sites were drilled using a (gas-powered) portable drill and oriented with both magnetic and solar compasses. Sites were chosen in order to allow a determination of bedding with sediment-volcanic contacts. The nature of the sediment (mainly shales) did not permit sampling of the baked zones for contact test. In order to test the effect of superficial alteration, one site (AD) was sampled in newly exposed rocks from a mining gallery. In the laboratory, each core was cut in standard (25 mm high) specimens for susceptibility and remanent magnetization measurements.

Microscopic observations of a representative set of thin sections reveal two lithologies. The first group is made of ancient basaltic flows (17 sites) having undergone an early spilitization process. Typical mineral assemblage are albite-chlorite-carbonates-opaques. Olivine ghosts and pyroxenes also appear in some cases. Opaque mineralogy is composed of rutile, pyrite and (titano-) magnetite. The second group (8 sites) corresponds to tuffs or pyroclastic breccia, originating from the

Copyright 1991 by the American Geophysical Union.

Paper number 90JB02226
0148-0227/91/90JB-02226$05.00

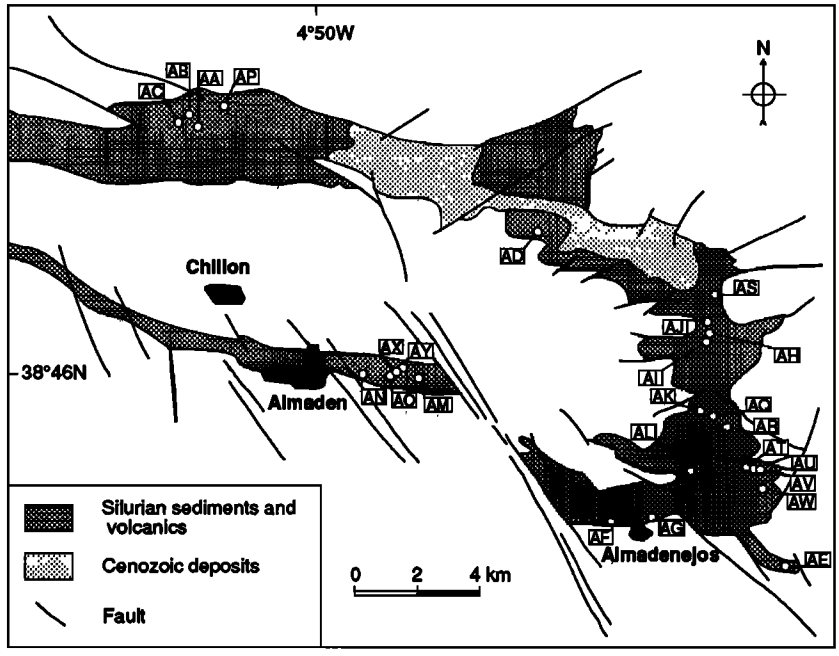


Fig. 1. Location map showing paleomagnetic sampling sites along the eastern termination of the Almaden syncline.

same magmatic stock, and commonly designated by the local term of "roca frailesca."

No evidence for metamorphism or deformation have been observed. From carbon and oxygen isotope analyses, *Eichman et al.* [1977] conclude that the igneous carbonates "represent genuinely hypogenic material derived from primary magmatic sources." However, a regional study of illite crystallinity and reflectivity of organic matter from shales [*Saupé et al.*, 1977] reveals that an anchizonal metamorphism has uniformly affected the Paleozoic formations.

Anisotropy of Magnetic Susceptibility

The purpose of studying the anisotropy of magnetic susceptibility (AMS) in such a paleomagnetic study is first to check if paleomagnetic directions could have been deviated by a strongly (>10%) anisotropic recording material, second to evaluate if Hercynian deformation has strained the rock and the preserved magnetizations, and eventually to control our bedding estimations from the nearby sediments. Furthermore, AMS may contribute to differentiate the magnetic phases present in the rock.

For AMS measurements, we have used a newly developed instrument composed of the electronic part of the standard Digico apparatus, interfaced with an Apple IIGS computer as described by *Tucholka and Tessier* [1987]. The performance of the Digico signal detection circuit has been improved by adding an amplifier/filter (Kronhite model 3321) before analog to digital conversion. This apparatus is driven by a new software in which original errors [*Veitch et al.*, 1983] of the Digico system were eliminated. Measurements of standard specimens [*Ellwood*, 1984] establish the validity of the overall procedure and calibration.

From each site, four to seven samples were measured. In 12 sites, no anisotropy (less than 1%) was detected. For the 13 remaining sites (mainly lavas), an anisotropy of susceptibility was present, which is quite consistent at the site level. Mean tensors are listed in Table 1, with anisotropy percentage, ratios of main susceptibilities and directions of maximum and minimum susceptibility axes. Anisotropy varies between 1 and 11%, with most sites in the range 3 to 7%. This is a clear but moderate anisotropy that can provide information concerning the history of the rocks but is not of such strength to significantly perturb the natural remanent magnetization (NRM) acquisition. Bulk susceptibility, main susceptibility ratios, and eccentricity of the susceptibility ellipsoïds vary substantially between sites (compare Table 1, Figure 2). These parameters could be linked, because many sites with high bulk susceptibility have susceptibility ellipsoïds which are prolate (eccentricity < 1) while many sites with low bulk susceptibility have susceptibility ellipsoïds which are oblate (eccentricity > 1) (Figure 2). This may reflect differences in the magnetic mineralogy between the flows.

The directions of the principal susceptibility axes are also somewhat smeared (Table 1 and Figure 3). In field coordinates (in situ), both minimum and maximum axes are distributed in a SW-NE broadband, which is perpendicular to the structural trend of the syncline; there is no clear grouping of either axis in this pattern. However, the orientation of this distribution, normal to the trend of the structures, suggests that the anisotropy predates folding and reflects emplacement rather than deformation features. Moreover, if strain were responsible for the AMS, we should expect a consistent distribution in field coordinates over the whole syncline. Furthermore, microscopic observations have not revealed any internal deformation in these rocks. We therefore conclude that the AMS was not originated during deformation, but probably earlier, and has been scattered by rigid rotations that occurred during the folding.

With the exception of only one site (AG), the bedding plane of each site contains (or at least is very close to) either the minimum or maximum susceptibility axis. There is therefore a relationship between the AMS tensor and the estimated paleohorizontal; this justifies using structural corrections to evaluate the initial (emplacement) anisotropy. The exception of site AG suggests that bedding is not well evaluated in this site,

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Anisotropy of Magnetic Susceptibility Results

Site	n	Susceptibility 10^{-5}SI	k_{max}/k_{int}	k_{int}/k_{min}	%An	Eccentricity	Directions of Principal Axes (declination/ inclination) k_{max} IS	k_{min} IS	k_{max} TC	k_{min} TC
AE	6	45	1.010	1.005	1.5	0.995	023/68	252/15	110/00	020/71
AF	3	1900	1.058	1.011	7.0	0.956	070/29	168/14	046/10	181/76
AG	5	1998	1.060	1.049	11.2	0.990	266/07	162/65	292/43	071/39
AK	5	68	1.011	1.007	1.8	0.996	037/28	298/17	211/17	111/31
AM	4	74	1.003	1.011	1.4	1.008	267/66	050/20	321/10	228/17
AN	5	429	1.006	1.049	5.5	1.043	243/30	031/57	285/43	020/06
AO	6	75	1.004	1.025	2.9	1.021	239/65	074/24	332/11	241/10
AR	4	2712	1.030	1.023	5.4	0.993	219/08	119/54	025/71	190/18
AS	5	173	1.013	1.060	7.4	1.046	171/06	073/55	177/12	335/78
AT	6	722	1.012	1.024	3.6	1.012	224/58	002/25	225/07	324/51
AW	7	720	1.059	1.013	7.3	0.957	205/77	064/11	225/02	128/77
AX	6	1319	1.030	1.014	4.4	0.984	086/23	327/49	250/06	157/28
AY1	3	2536	1.043	1.013	5.7	0.971	080/21	239/68	245/05	336/12
AY2	2	84	1.003	1.008	1.1	1.005	196/53	070/24	342/37	241/14

The number of samples is n, k_{max}, k_{int}, k_{min} are the principal susceptibilities, $\%An$ is the anisotropy percentage. IS and TC mean in situ (field coordinates) and tilt-corrected (bedding coordinates), respectively.

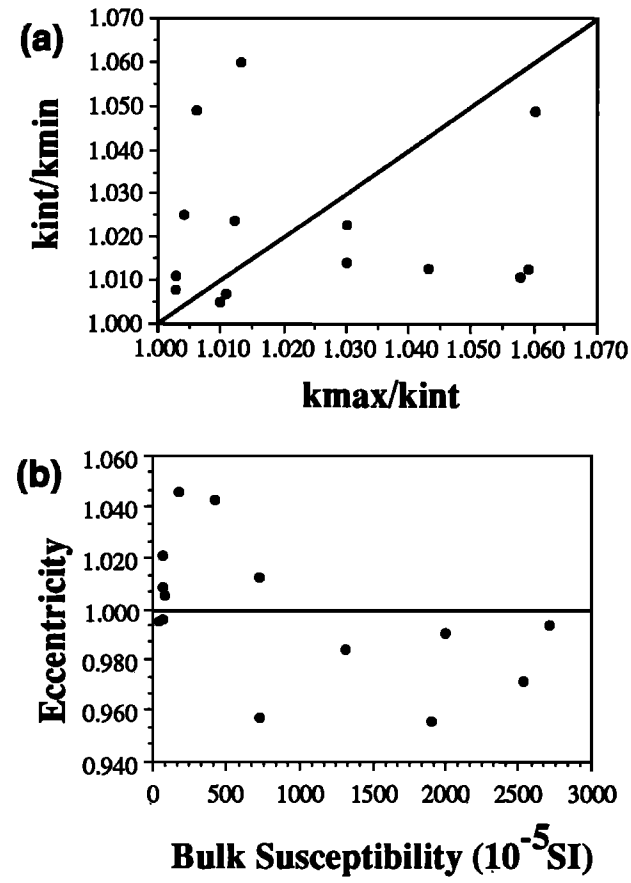


Fig. 2. Site-mean susceptibility parameters of sites with anisotropy > 1%: (*a*) Flinn diagram of main susceptibility ratios; (*b*) eccentricity versus bulk susceptibility. In both diagrams the solid line separates the prolate and oblate fields.

which is probably the case because we have not been able to find a clear sedimentary contact for this flow in the field. For all other sites, the agreement between bedding and AMS validates the bedding determinations, which are always difficult to obtain in volcanic rocks.

Bedding corrections were therefore applied to the AMS principal axes (Figure 3). As a result, the pattern of susceptibility axes divides in two sets, one being more or less horizontal, with NE-SW azimuths, the other in a NW-SE vertical plane. Minimum and maximum axes are present in each set, which confirms that the AMS is due to an inhomogeneous magnetic mineralogy. While most of the sites show good within-site agreement of the susceptibility parameters, an exception is site AY. The detailed investigation of AMS results from this site permits us to show that the measured AMS is due to the combined effects of two kinds of magnetic minerals, the proportions of which defining the overall signature of each sample. In this flow we can observe two groups of samples (compare Table 1): group AY1, from the middle part of the flow, reveals a high susceptibility, high anisotropy and very prolate ellipsoïds; group AY2, from the bottom of the flow, has a low susceptibility, low anisotropy and oblate ellipsoïds; finally, the maximum axis from AY1 is parallel to the minimum axis of AY2. Under the microscope, we have observed in the two groups small distributed cubic magnetite grains, of slightly lower size in the flow border (AY2). However, large elongated magnetite grains, of apparently primary origin

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

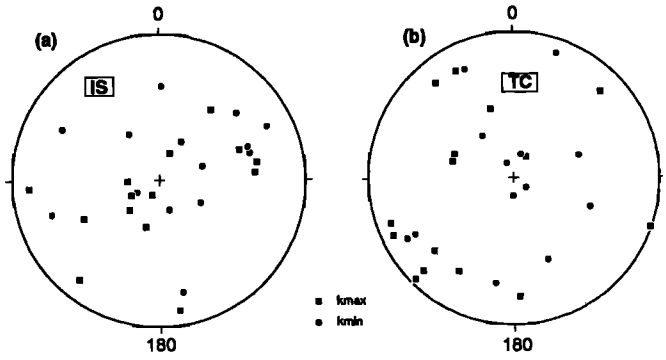


Fig. 3. Stereographic projections of the site-mean directions of the main susceptibility axes, (*a*) before and (*b*) after simple tilt correction.

(showing reactional margins and cooling cracks) are present only in the middle part of the flow. The very anisotropic shape of these grains will result in a high anisotropy, if they have a preferred orientation, due to the flow of the lava. This anisotropy seems to dominate the AMS of group AY1. In the bottom of the flow, the more rapid cooling may have prevented the crystallization of similar grains, and the observed magnetic foliation (perpendicular to the flow of the lava) is probably associated with the distributed small magnetite grains.

The distinction between the AMS from AY1 and AY2 is representative of the differences between sites in the rest of the AMS results. Eight sites (plus probably AG, excluded because of its bedding uncertainty) have well-determined k_{max} generally prolate ellipsoïds, and often high susceptibility. Their maximum axes cluster around a shallow NE-SW mean direction (231/+06); corresponding minimum axes are distributed in a NW-SE vertical plane (Figure 4). The five remaining sites have generally lower susceptibility and oblate ellipsoïds. Their minimum axes cluster around the same shallow SW direction (220/+11), and the maximum axes again define a NW-SE vertical plane (Figure 4). As a conclusion, it appears that according to the changing magnetic mineralogy we have a complete inversion of the AMS tensors between these two groups of flows.

Tilt-corrected AMS axes are still somewhat smeared, especially for the k_{max} cluster (Figure 4). However, it is important to note that the Almaden syncline is a quite complex structure, with steeply plunging fold axes, and thus the simple tilt correction is not sufficient to restore the flows in their original positions. The application of such correction is useful to restore the inclinations of the AMS axes but induces an error in the estimated declinations. It is quite significant in this respect that the dispersion in the corrected AMS axes resides mainly in the declinations, while inclinations are very consistent. It is not clear, however, which part of the dispersion is due to the initial scatter of the AMS axes, and which is due to the correction procedure. This prevents us from using the AMS axes to define better correction algorithms for rigid deformations.

From the detailed analyses of the AMS measurements, we conclude that these flows present a magnetic anisotropy reflecting the conditions of their emplacement, probably from northeast, on a gentle slope (<10°). The AMS tensors are inverted between flows, according to the relative quantity of the different magnetite phases. The deformation of the flows during the Variscan orogeny have disturbed the AMS axes directions (especially the declinations), preventing a more precise localization of the volcanic centers.

PALEOMAGNETIC ANALYSIS

Remanent magnetization measurements were done with either the Schonstedt DSM-1 spinner magnetometer at Rennes, or cryogenic magnetometers at IPG Paris (CTF) or Rennes (LETI). Thermal demagnetizations were conducted at Rennes with the Schonstedt TSD-1, and at Paris with a laboratory-made large-capacity furnace. A Schonstedt GSD-1 alternating fields (af) demagnetizer was available in both laboratories. Demagnetization trajectories were interpreted by fitting straight lines through the data points, with the help of an interactive computer-based graphical system, as described by *Perroud* [1985]. Thermomagnetic experiments were also conducted with an automated Brucker Curie balance at Rennes.

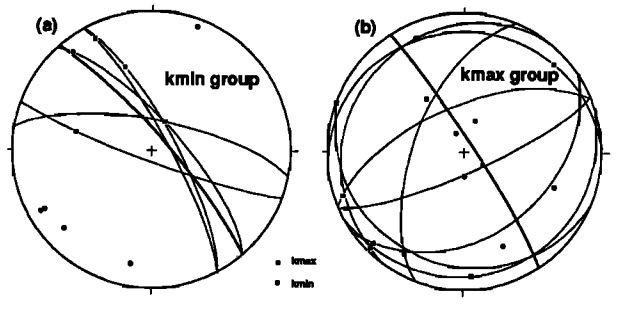


Fig. 4. Tilt-corrected magnetic lineations (square) and foliations (thin great circles; solid circles are foliation poles; thick great circles are the mean magnetic foliations) for the two groups of sites: (*a*) sites with low susceptibility and oblate ellipsoid, showing k_{min} clustered around a NE-SW horizontal direction; (*b*) sites with high susceptibility and prolate ellipsoid, showing k_{max} clustered around the same NE-SW horizontal direction.

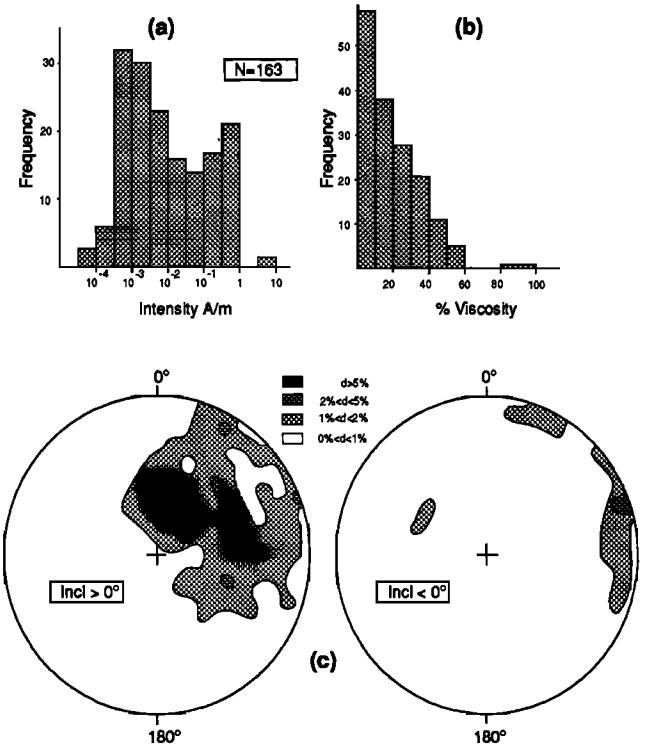


Fig. 5. NRM results for a selection of one specimen from each core: (*a*) intensity, (*b*) viscosity, (*c*) direction density contour. Density of observations per unit area is represented by *d*.

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

A short period viscosity test [*Thellier and Thellier*, 1959] was conducted before any demagnetization. It consists in storing the specimens orientated in the laboratory field during a short duration (typically 10 days), then measure NRM, store again in reverse orientation for the same duration and measure again; this allows the determination of the true remanent magnetization intensity, direction (free of short-term viscosity), and the viscosity index (ratio of intensities of viscous and remanent magnetizations), which is a helpful indicator for the interpretation of the soft components of magnetization. One specimen from most core was analyzed (163). Results are displayed in Figure 5, in the form of a density contour plot for the directions and of bar diagrams for the NRM intensities and viscosity indexes.

NRM intensities show a rather large dispersion, between $5x10^{-5}$ and 5 A/m, with two modes at 10^{-3} and 0.5 A/m. Flows generally fall into the high-intensity group, with some exceptions. All volcano-sedimentary rocks belong to the low intensity group. Viscosity indices are distributed between 0 and 50%, with the majority (60%) being below 20%. Once again, a distinction is possible with respect to rock type, since most flows displayed low viscosity indices (less than 20%), while tuffs and breccias had viscosity indices between 20% and 50%. NRM directions appear highly dispersed, with a strong bias toward the dipolar present-day Earth magnetic field (EMF) at Almaden (D= 0°, I= +58°). However, other directional trends exist, one being east with intermediate-downward inclination, and the other northeast with shallow-upward inclination. As a conclusion of this test, we can anticipate that the remanent magnetization has a composite nature, which requires careful specimen-by-specimen progressive demagnetizations rather than optimum cleaning methods. According to their different magnetic properties, the results of the demagnetization procedures for the flows and the breccia will be discussed separately.

Demagnetization Results From the Tuffs

Three components of magnetization were revealed at the eight breccia sites (Figure 6), upon thermal or af treatments. The first removed was found in all sites and disappears usually by 150°-200°C or 20 mT. The direction for this component in field coordinates is similar to the present-day EMF direction, and it is dominant in some sites (compare sample AB12, Figure 6) with a high-alteration state. It was therefore interpreted as a recent overprint due to in situ alteration processes. Above 200°C, most sites display a shallow-eastward component, which can dominate the rest of the NRM (sample AE29, Figure 6), but in all cases the demagnetization trajectories do not converge toward the origin. By 450°C or 50 mT, final northeast and slightly upward directions are reached for some samples from sites AD,AJ and AK (Figure 6). For site AI, similar endpoints were observed, but with shallow-westward directions. The accurate determination of the directions of these endpoints was extremely difficult due to very strong viscous components appearing in the samples after heating at 500°C; thermomagnetic cycles made on such samples (Figure 7, AD23) confirm that chemical alteration resulted from heating above 400°C, with large quantities of magnetite being produced (Curie point at 580°C). Furthermore, af demagnetization usually produced unstable demagnetization behavior. When both af and thermal treaments were successful up to 500°C or 60 mT (site AK, Figure 6), they give very consistent results.

Demagnetization Results From the Lavas

In the lava samples a present-day field component can also be observed, although it is not a general case. When present, it is easily removed by heating to 200°C or with fields of 20 mT (compare AG43 or AY63, Figure 8). An intermediate east-southeast and shallow component appears in most sites (AG43, Figure 8). After removal of the intermediate component, magnetization directions remain stable while intensities decay progressively up to 580°C or 100 mT. The directions of these endpoints can be very different in field coordinates (in situ); a large number are in the northeast quadrant, with shallow to moderate upward inclinations, but others have steep downward inclinations (compare AR12 or AV40, Figure 8).

On top of these general features, some peculiarities were observed (Figure 9): (1) strong secondary components which have variable directions and which are not aligned with the present-day field appear in some sites (AX59 or AF42, Figure 9); they resist heating up to 500°C, but are quickly demagnetized with fields of 20 mT; however, after their elimination, the usual stable endpoints are reached; (2) some samples display complex demagnetization trajectories (AW55, Figure 9) or insufficient separation of components (AY66, Figure 9); (3) most samples reveal maximum unblocking temperatures in the range 500°-580°C, but in some cases unblocking temperatures below 350°C can be observed (AM92, Figure 9). Thermomagnetic experiments on a representative set of samples confirm that magnetite is the main magnetic carrier (Figure 7): high susceptibility samples show very clear Curie points at 580°C, with nearly reversible behavior (AF42, AR14, Figure 7); similar Curie points were obtained on low-susceptibility samples, but it was necessary to extract the magnetic phase from the rock to get a clear result (AV43, Figure 7). However, there are indications for another magnetic phase that loses its magnetization between 300° and 350°C, as revealed by sample AM91, or by the difference between the heating and cooling curves for AR14. This second phase (maghemite?) is probably carrying the NRM of site AM samples since they have very similar unblocking temperatures (Figure 8).

The comparison of AMS and paleomagnetic results reveal clear relationships between them. In site AY, for example, the high-temperature components were obtained mainly in the high-susceptibility samples (compare AY63, Figure 8), while separation of components was not successful in the low-susceptibility samples (compare AY66, Figure 9). Intensity of magnetization is also very much higher in the former. It seems therefore that the large primary magnetite grains observed in the high-susceptibility samples, which present reversible thermomagnetic behavior and Curie points at 580°C, are actually carrying the high-temperature component. As they have recorded emplacement features in the AMS, we believe that they have originated very early in the crystallization process, and they can therefore carry the primary magnetization of the Almaden lavas.

DISCUSSION OF THE PALEOMAGNETIC RESULTS

Site-mean directions of both the intermediate and final components of magnetization are summarized in Table 2 for the tuffs and Table 3 for the lavas.

Most of the tuff sites reveal between 200° and 450°C an east-southeast shallow component (Figure 10). One noticeable exception is site AD, sampled within an active mining gallery. After tilt correction, the scatter increases (Fisher parameter k decreases by a factor of 3), and since inclinations became very

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

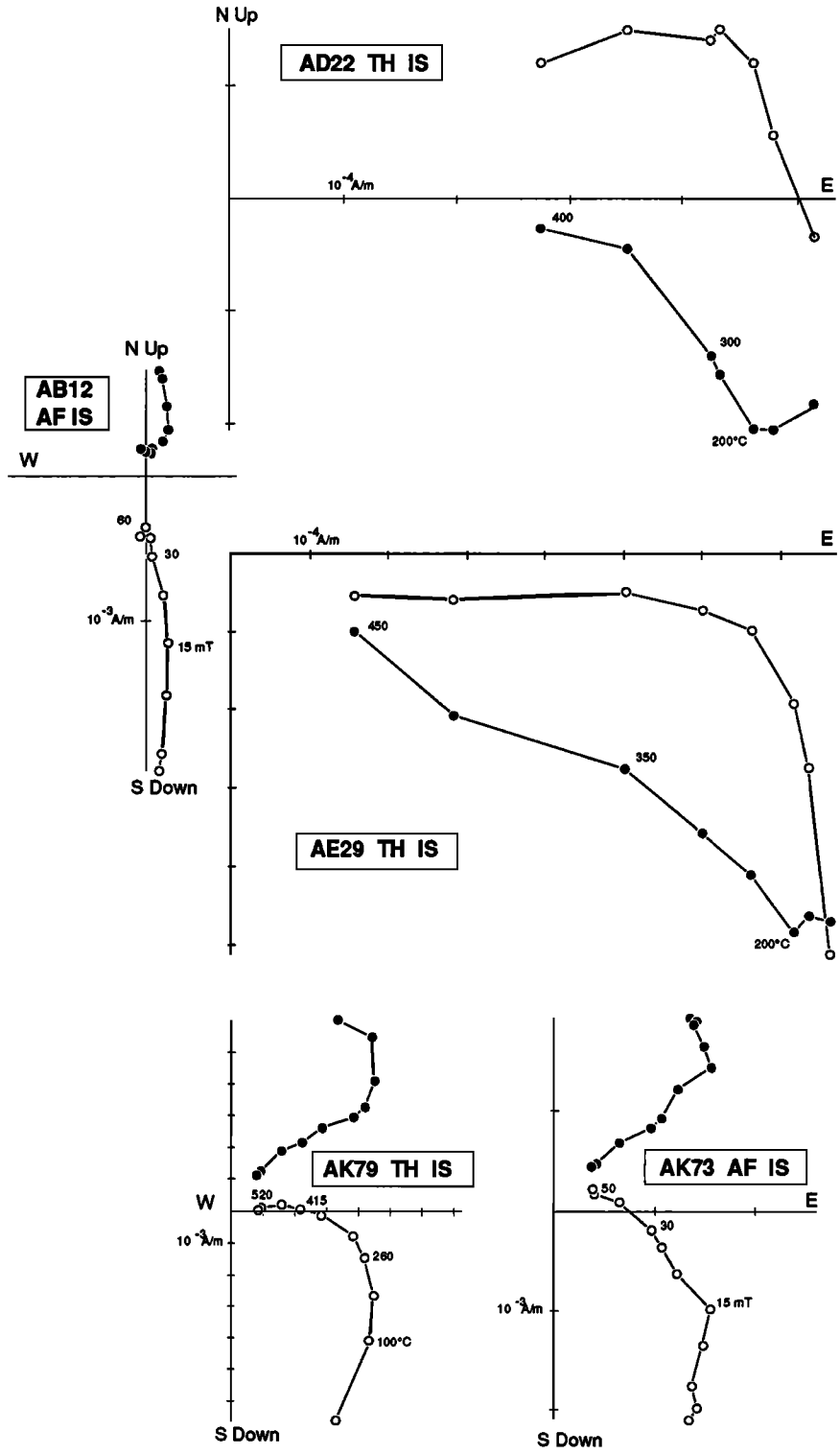


Fig. 6. Orthogonal projections of selected demagnetization curves for the tuffs, in field coordinates. Closed (open) circles represent the projection onto the horizontal (vertical) plane. NRM intensities are given in ampere per meter, temperature in degrees Celsius and af peak value in millitesla.

dispersed, more sophisticated unfolding algorithms will not improve the situation. It is clear therefore that this component is postfolding. Its in situ mean direction is D=104°, I=+21°, α_{95}=20°.

Characteristic components could be isolated in most samples at sites AD and AK or in certain samples from other sites. Because of the small total number of characteristic directions obtained from the tuffs (14), site-mean directions are not always significant. Nevertheless, individual sample directions are plotted in Figure 11, both before and after tilt correction. Most are in the northeast (upward) quadrant, but three are in the opposite southwest quadrant. Possibly, two polarities were

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

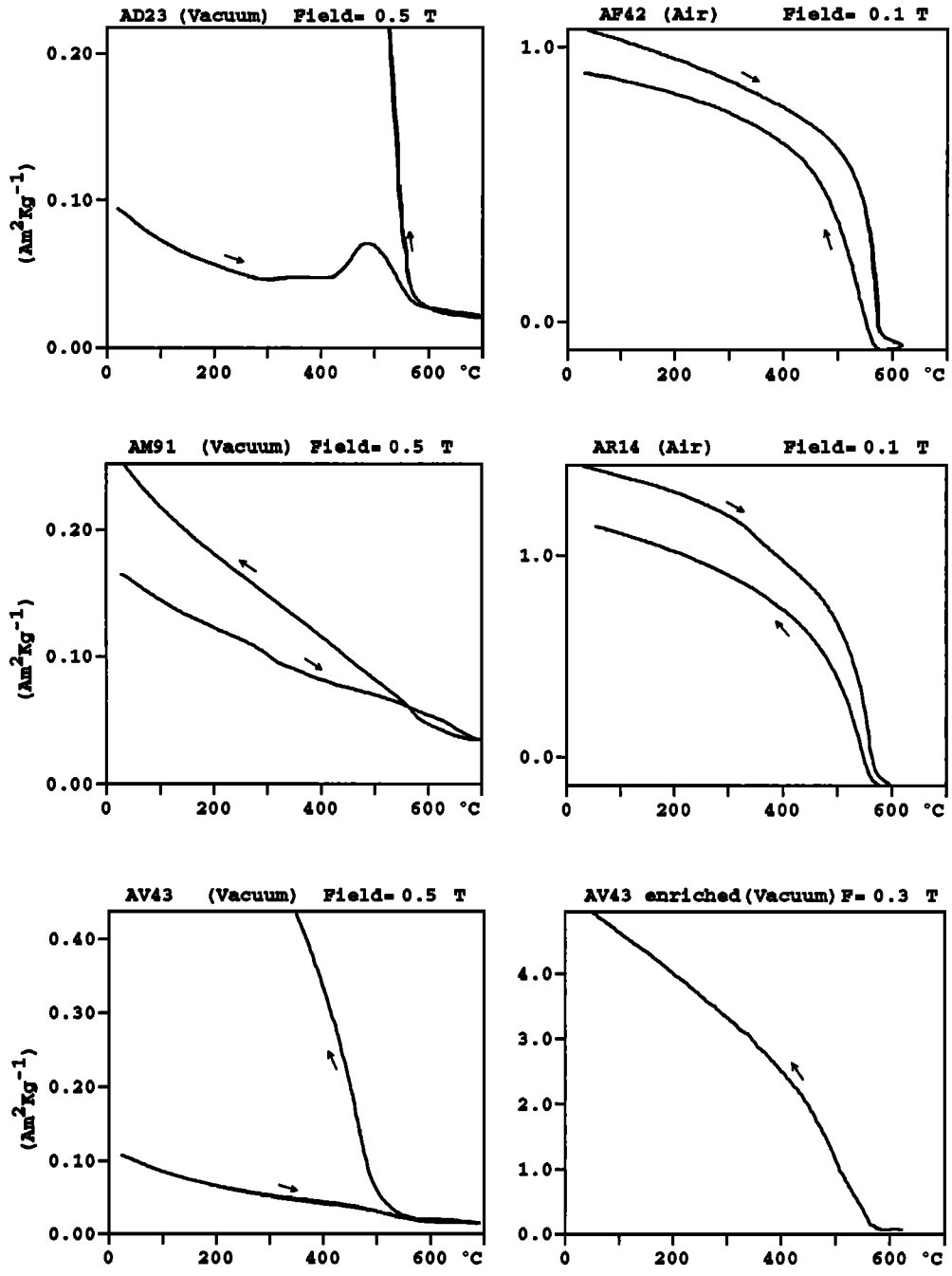


Fig. 7. Thermomagnetic experiments for a representative set of samples. Applied fields are expressed in tesla.

recorded, although directions are not exactly antiparallel, nor in situ neither after tilt correction. Given the difficulties mentioned above with respect to isolation of characteristic directions in the tuff samples, we suspect that isolation of the individual components was not completely successful, with the exception of samples from site AK (compare Figure 6). Subsequently, we will verify that site AK is the only tuff site that gives results consistent with those of the lavas.

From the lavas, 13 out of 17 sites reveal the east-southeast shallow components, in the temperature range 200°-450°C (Figure 10). Upon tilt correction, the substantial increase of dispersion (k decreases from 22 to 8) indicates again a conclusive negative fold test. The in situ mean direction of the lavas (D=111°, I=+06°, $\alpha 95$=9°) is identical to that for the tuffs. Thus both rock types have simultaneously (after folding) undergone a remagnetization event which has partially overprinted the remanent magnetization. The absence of this remagnetization in site AD, from unaltered fresh rock, suggests that alteration might have played a role for its acquisition, in the same way that present-day surface alteration seems to control present-day field components. Considering the 20 sites where it has been identified together, the mean remagnetization direction is D=109°, I=+11°, $\alpha 95$=9°. It presents a 99% statistically significant negative foldtest (k ratio is equal to 2.8 for 20 entries) and corresponds to a paleomagnetic pole at 11°S, 69°E (dp=5°, dm=9°) and a paleolatitude of 6°S.

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

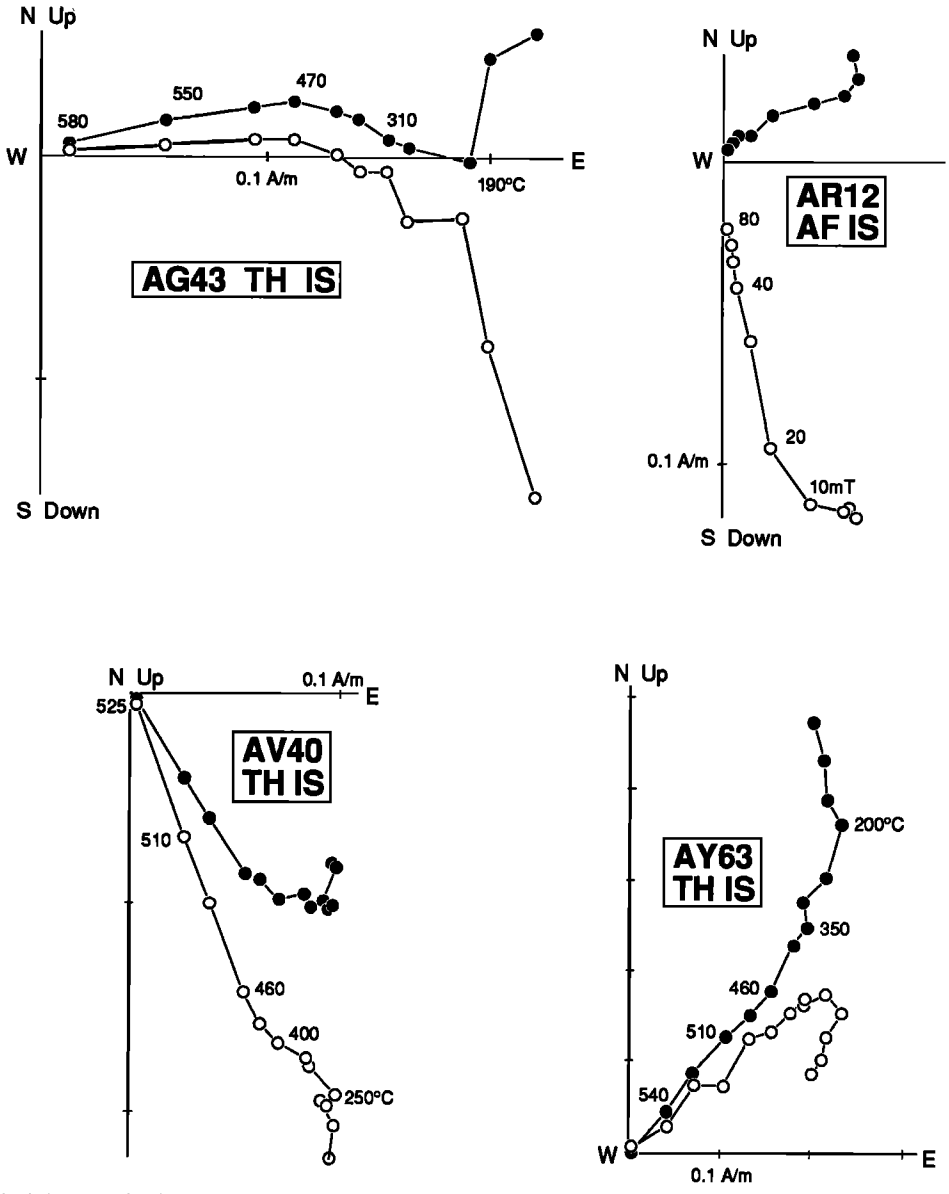


Fig. 8. Typical demagnetizations curves for the lavas showing well-isolated high-temperature components with quite different directions in field coordinates (same conventions as Figure 6).

After elimination of this remagnetization, characteristic components with unblocking temperatures up to 580°C were found in all lava sites but AC (Figure 11). At site AL, which contains a very strong present-day field overprint, only two samples reveal the high unblocking temperature component. At site AV, these components fall into two nearly antiparallel groups, which indicates that both polarities were recorded. Finally, three sites (AT, AU and AW) display elongated distributions of the characteristic directions (compare site AW, Figure 11), resulting in very poor statistical parameters for the Fisher site-mean ($5<k<11$). The origin of such distributions is not clear and those sites were not used for the final mean calculations.

The overall in situ distribution of site-mean characteristic directions (Figure 11) shows a large scatter; both declination and inclination are dispersed. If we rejected problematic sites, such as those mentioned above, the situation is not much improved. The application of the simple tilt correction does not lead to a real improvement of the dispersion. However, it reveals that three sites (AR, AS, AV) represent directions of the opposite polarity. Moreover, the site-mean inclinations are now quite consistent, after choosing uniform polarities for all the sites, (mean -39°, standard deviation 17°), as revealed by Figure 12, which is a plot of inclination versus declination (problematic sites AL, AT, AV and AW are not plotted in Figure 12). This is strong evidence for the prefolding origin of this magnetization. The residual declination dispersion is not surprising in the complex tectonic context of the Almaden syncline. To better estimate the prefolding directions of magnetization, more sophisticated unfolding algorithms are needed, taking into account the plunge of the fold axis (60° to 304°). We have used here a two step unfolding method: the first step consists in rotating the fold axis to the horizontal; the second completes the unfolding, using the residual bedding attitude. The application of such a method to our data leads to a very clear decrease in the declination dispersion (Figure 12), and we therefore conclude that (1) the

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

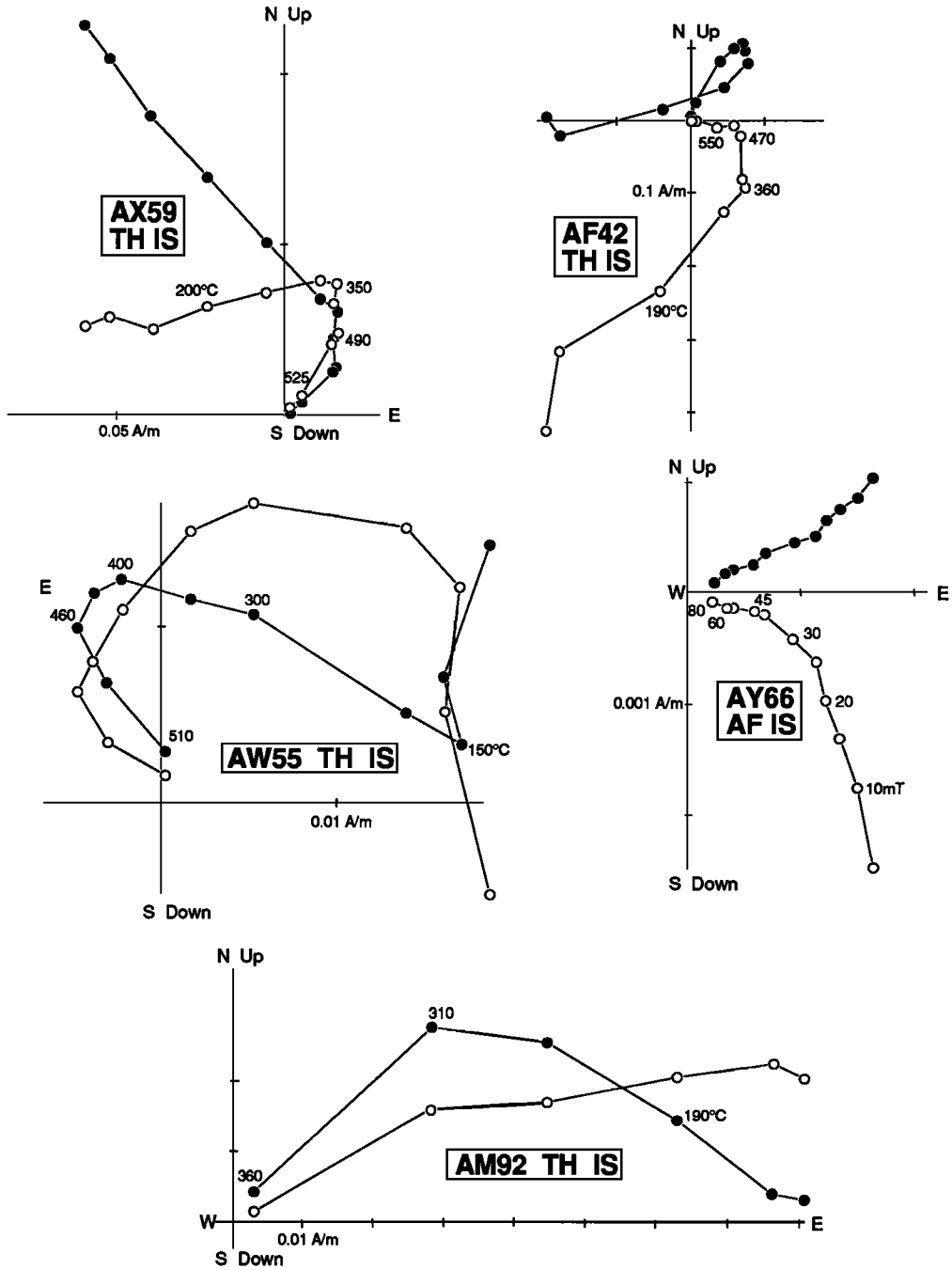


Fig. 9. Selected demagnetizations curves for the lavas, illustrating the different peculiarities described in the text (same conventions as Figure 6).

magnetization is prefolding, and (2) unfolding has to take into account the plunge of the fold axes.

The best estimate of the paleomagnetic mean direction from these lavas was obtained by selecting the best sites, and by applying the unfolding method described above. Site AG was not used because of its structural attitude uncertainty (compare above). Sites with k lower than 15 or $\alpha 95$ greater than 20° or with problematic separation of components were rejected. Site AK (tuffs) which fits these reliability criteria and presents a very similar direction, was added to the 10 residual lava sites. The mean direction obtained using these 11 sites is D=62°, I=-36°, $\alpha 95$=14°. Its corresponds to the pole position latitude 8°N, longitude 118°E, dp=9°, dm=13°, and a paleolatitude of $\pm$20°, according to the polarity choice (normal or reverse). Using the same 11 sites, the Fisher dispersion parameter, k, is 5 in field coordinates, 5 after tilt correction and 12 after unfolding. Unfolding thus leads to a 95% statistically significant clustering of the site-mean directions (k ratio of 2.4 for 11 entries).

INTERPRETATION OF THE PALEOMAGNETIC COMPONENTS

Overprint

The low inclination and the reverse polarity of all sites suggest that acquisition took place in the Late Paleozoic reverse polarity interval (Kiaman), as in many other remagnetization

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Paleomagnetic Results From the Tuffs

Site	*n/ns*	Th Range	Af Range	Dec_{IS}	Inc_{IS}	*k*	α_{95}	Dec_{TC}	Inc_{TC}
AA	5/8	200°-450°C		150	+26	35	13	150	+06
AB	3/7	300°-450°C		097	+28	70	15	132	+14
AD	5/8	350°-450°C	30-50 mT	086	-25	69	09	075	+06
AE	7/9	250°-450°C	25-50 mT	114	+06	39	10	211	+62
AH	4/6	200°-500°C	30-100 mT	095	+24	13	27	145	+57
AI	4/8	300°-450°C	40-60 mT	106	+07	180	07	099	+44
	3/8	450°C		263	-15	12	38	246	-42
AJ	5/8	250°-400°C	20-50 mT	101	+18	128	07	097	-22
	2/8	400°-500°C		076	-09	-	-	072	+05
AK	6/8	300°-450°C	20-50 mT	063	+27	38	11	044	+03
	4/8	450°-530°C	50-60 mT	034	-11	21	20	068	-39
Mean IS	7	200°-450°C	20-50 mT	104	+21	10	20		
Mean TC	7	200°-450°C	20-50 mT			3	46	117	+31

The number of entry in the Fisher statistics is *n* and *ns* is the total number of samples analyzed. Th and Af ranges denote the experimental interval in which the magnetizations were determined; *k* and α_{95} are the Fisher statistical parameters. Calculated means concern the intermediate component found in all sites but AD.

TABLE 3. Paleomagnetic Results From the Lavas

Site	*n/ns*	Th Range	Af Range	Dec_{IS}	Inc_{IS}	*k*	α_{95}	Dec_{TC}	Inc_{TC}	Dec_{UF}	Inc_{UF}
				Characteristic Components of Magnetization For All Sites							
AC	8/10	300°-450°C	20-40 mT	113	+26	23	12	123	+09		
AF	9/12	300°-580°C	30-50 mT	028	-01	56	07	050	-40	030	-40
AG	7/9	100°-400°C	10-40 mT	117	+18	14	17	109	-04		
	9/9	400°-580°C	30-60 mT	058	-09	25	11	080	-65	042	-65
AL	2/8	250°-400°C	15-30 mT	056	-22	-	-	056	+11		
AM	7/10	100°-300°C	0-15 mT	127	-02	70	07	097	+48		
	10/10	300°-400°C	15-30 mT	041	-14	48	07	081	-36	057	-36
AN	8/9	250°-500°C	20-40 mT	109	+04	24	12	101	+12		
	3/9	510°-540°C	40-60 mT	077	-23	215	08	105	-29	079	-29
AO	6/8	200°-350°C	0-50 mT	035	+01	35	12	083	-50	047	-50
AP	6/6	250°-400°C	20-40 mT	121	-04	18	16	102	-41		
	3/6	400°-500°C	40-60 mT	086	-55	14	35	026	-36	053	-36
AQ	6/9	250°-400°C	20-50 mT	118	+11	20	15	109	+41		
	7/9	400°-520°C	50-70 mT	058	-35	15	16	079	-27	073	-27
AR	3/8	300°-550°C	10-30 mT	119	-01	19	29	133	+17		
	5/8	300°-550°C	30-80 mT	058	+70	82	09	221	+30	268	+30
AS	4/10	150°-450°C	20-40 mT	123	-01	29	18	129	+31		
	4/10	450°-500°C	10-30 mT	099	+65	11	30	261	+75	272	+75
AT	5/11	300°-450°C	10-40 mT	104	-01	17	19	114	+23		
	8/11	300°-570°C	20-50 mT	092	-48	05	29	075	-06		
AU	4/12	200°-400°C		117	+23	14	25	143	+29		
	10/12	350°-550°C	30-70 mT	047	-15	07	19	047	+37		
AV	3/8	250°-450°C		091	-15	290	07	095	+25		
	4/8	450°-550°C	10-30 mT	142	+54	81	10	188	+28	217	+28
	3/8	250°-500°C	30-50 mT	354	-44	44	19	013	-06	042	-06
AW	7/11	200°-450°C	0-20 mT	104	+11	21	14	144	+38		
	9/11	400°-550°C	20-80 mT	068	-50	11	16	063	+24		
AX	5/9	400°-550°C	10-40 mT	028	-40	36	13	156	-48	116	-48
AY	4/13	200°-450°C	10-60 mT	082	+01	17	23	085	-05		
	7/13	300°-550°C	10-30 mT	049	-24	52	08	114	-35	078	-35
				Mean Overprint							
	13			111	+06	22	09				
	13					08	16	114	+19		
				Mean Characteristic Magnetization From Selected Sites (k > 15 and α_{95} < 20°, Including AK)							
	11			034	-34	05	22				
	11					05	22	069	-41		
	11					12	14			062	-36

Same conventions as Table 2; in addition, unfolded (UF) directions are given for the characteristic magnetization,obtained by first rotating back the fold axis to horizontal and then correcting for residual tilt.

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

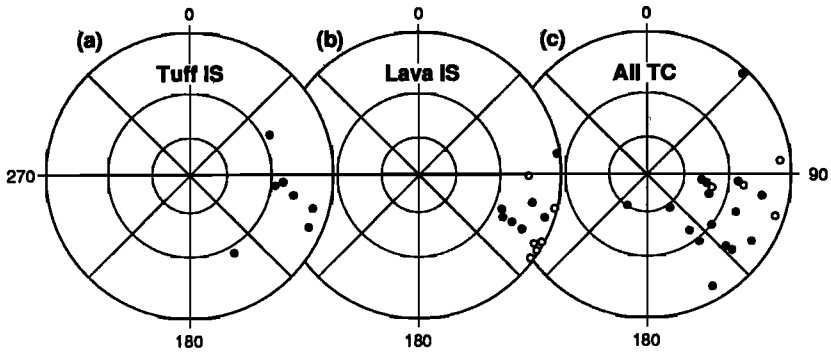


Fig. 10. Stereographic projections of the site-mean directions of the Late Paleozoic overprint, showing the similarity of the results from (*a*) the tuffs and (*b*) the lavas and (c) the postfolding origin of that component. Closed (open) circles represent projections onto the lower (upper) hemisphere.

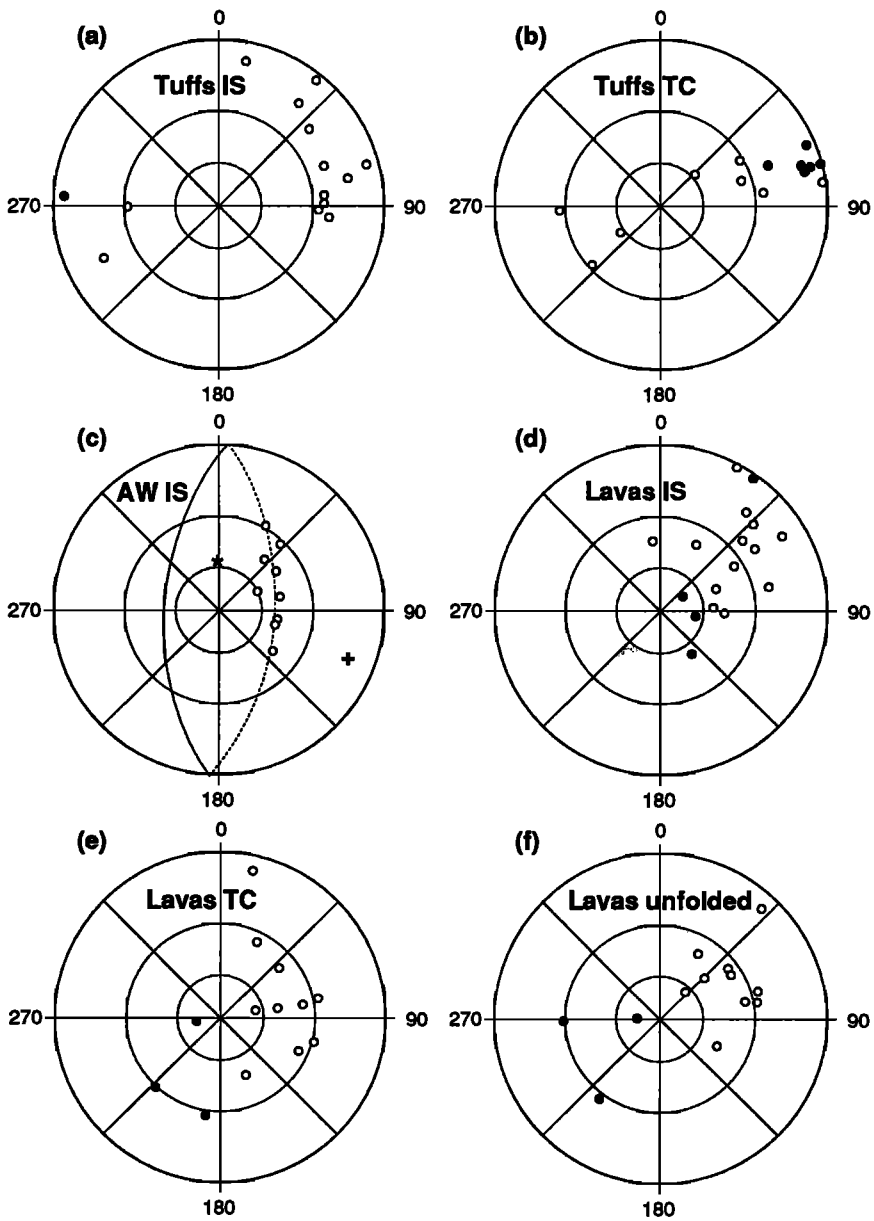


Fig. 11. High-temperature components: (*a*, *b*) sample directions obtained from the tuffs before and after tilt correction; (*c*) great circle distribution of samples directions from lava site AW (the asterisk represents the present-day EMF, the plus the Late Paleozoic overprint); (*d*) in situ site-mean directions for all lava sites; (*e*, *f*) tilt-corrected and unfolded site-mean directions of lava sites, after rejection of unreliable results (same conventions as Figure 10). Unfolding consists in rotating back the fold axis to horizontal, then correcting for the residual tilt.

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

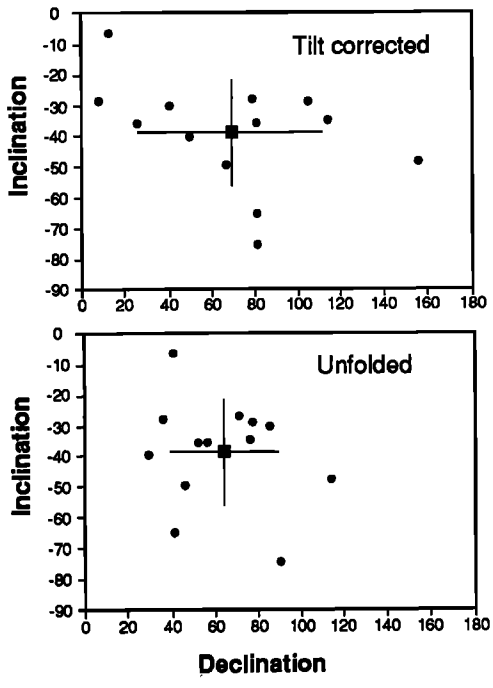


Fig. 12. Comparison of the declination dispersion for the two kinds of structural corrections applied to the high-temperature components. Squares represent the mean, and the thin lines the standard deviations.

events. However, the declination is significantly different from the Late Paleozoic expected direction in Spain (D=155,° I=+10; *Van der Voo* [1969]). This 46° counterclockwise rotation can be related to the tectonic closure of the Ibero-Armorican arc [*Perroud and Bonhommet*, 1981]. It provides further evidence that rotations have occurred in western Europe after the main folding phases of the Variscan orogeny [*Perroud*, 1986].

High Temperature Component

According to the fold test, this component was acquired before the Variscan deformation. As these basaltic flows do not reveal any metamorphism or internal deformation, and the analysis of the magnetic mineralogy has shown that primary magnetites are actually carrying that component, we propose that the observed mean direction is a reliable record of the Silurian paleomagnetic field. The AMS study has shown that anistropy is weak but reflects emplacement on a gentle slope (<10°) from northeast. We can therefore exclude a bias of the magnetization direction due to anisotropic material or pronounced dipping of underlying rocks. The presence of both polarities indicates acquisition of magnetization occurred over a sufficiently long period to average out secular variation of the EMF; the presence of antiparallel directions indicates that characteristic components were suitably isolated.

A critical question concerns the normal or reverse polarity of the mean direction (D=62°, I=-36°), and the estimation of the paleolatitude. Due to its northern declination, it is reasonable that this direction is of normal polarity and would correspond to a paleolatitude of 20°S for the study area. This interpretation is quite reasonable, since Ordovician to Devonian paleogeographic reconstructions [*Perroud et al.*, 1984; *Van der Voo*, 1988; *Paris and Robardet*, 1990] place Spain in the southern hemisphere. However, the observed Silurian magnetization has to be corrected from younger tectonic events such as, (1) the Cretaceous opening of the Bay of Biscaye [*Van der Voo*, 1969] and (2) the Variscan closure of the Ibero-Armorican arc [*Perroud and Bonhommet*, 1981]. To take into account event 1, it is necessary to rotate 35° clockwise the observed declination [*Van der Voo*, 1969]. For event 2, we have seen above from the direction of the Late Paleozoic overprint that the magnetizations must be rotated about 46° clockwise. As a result of these corrections, a mean direction of D=143°, I=-36° would clearly point southward. Thus unless a very large additional rotation is postulated, a reverse polarity must be assumed for the characteristic magnetization of the Almaden volcanics, and therefore a paleolatitude of 20°N. As the Almaden syncline belongs to the central part of the Ibero-Armorican arc, which has constituted a coherent unit in the whole Paleozoic [*Paris and Robardet*, 1977], and since Ordovician paleomagnetic results from this geological unit (see below), although quite steep, have not indicated such a rotation, it is reasonable to exclude the possibility that a very large post-Silurian and pre-Variscan rotation was responsible for the southern corrected declination.

Geodynamic Implications

As Spain has not behaved as a rigid continental unit since the Lower Paleozoic [*Perroud and Bonhommet*, 1981], an apparent polar wander path (APWP) cannot be easily drawn from the paleomagnetic results. However, the inclination of the remanent magnetizations is not disturbed by vertical axis rotations like those involved in the Ibero-Armorican arc tightening or the Bay of Biscaye opening. Thus it is still possible to derive paleogeographic considerations from the paleolatitude estimates. We have therefore compiled previous Paleozoic paleomagnetic results from Spain, North of the Badajoz-Cordoba suture zone (Table 4), and established a paleolatitude versus time curve (Figure 13) that we shall compare to Gondwana's, recomputed for a site situated on the margin of the West African craton [*Bachtadse and Briden*, 1990]. Results from southernmost Spain were not used since it was probably accreted to Spain only in Late Paleozoic. On the other hand, some results previously not well understood and interpreted as secondary remagnetizations were reinterpreted in the light of the new Almaden result. This is the case for the Late Ordovician dolerites from Buçaco [*Perroud*, 1982] or for the Ordovician volcanism at Cabo de Penas [*Perroud*, 1983], where Lower Ordovician lavas have steep inclinations while Upper Ordovician lavas yield shallow inclinations. Our first interpretation was to consider the shallow magnetizations to be the result of Late Paleozoic overprinting. This is not

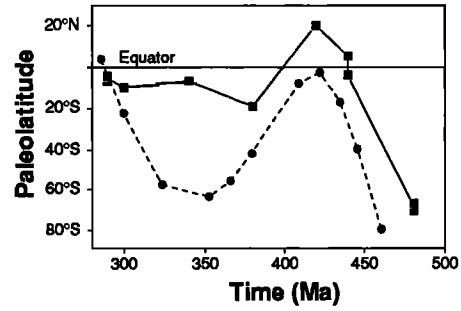


Fig. 13. Paleolatitude versus time curves for northern Spain (solid curve, data from Table 4) and the northern margin of Africa (dashed curve, after *Bachtadse and Briden* [1990]), from Ordovician to Carboniferous.

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. Paleozoic Paleomagnetic Results From Spain

Formation	Age, Ma	n	Declination	Inclination	α95	Paleolatitude	Reference
Redbeds Buçaco	480	1	151	+80	06	70°S	*Perroud* [1982]
Volcanism Cabo Penas	480	1	202	+78	06	67°S	*Perroud* [1983]
Volcanism Cabo Penas	440	1	160	+07	04	04°S	*Perroud* [1983]
Dolerites Buçaco	440	2	103	-09		05°N	*Perroud* [1982]
Spillites Almaden	420	11	062	-36	14	20°N	this work
Redbeds Cabrillanes	380	8	113	+34	10	19°S	*Perroud* [1984]
Redgrits San Emiliano	340	6	102	+13	02	07°S	*Bonhommet et al.* [1981]
Redbeds Cabo Penas	300	8	178	+19	09	10°S	*Perroud* [1983]
Overprint Almaden	290	20	109	+11	09	06°S	this work
Mean Permo-Carboniferous	290		155	+10		05°S	*Van der Voo* [1969]

The number of sites where the magnetizations were observed is n. For paleolatitude estimations, we have chosen polarities which minimize vertical axis rotations.

necessary any more since a Late Ordovician acquisition age is perfectly concordant with the Lower Ordovician - Silurian segment of the paleolatitude curve (Figure 13) drawn from published Lower Ordovician results [*Perroud and Bonhommet*, 1981] and our new Almaden result. We certainly cannot consider these shallow magnetizations as fully reliable and well-dated paleomagnetic data; we want just to stress that they support the contention that Armorica might have come througth the equator in as early as the Upper Ordovician.

The most striking feature of figure 13 is the resemblance of the Spanish and Gondwana paleolatitude curves in the interval 480-380 Ma, even if there exists a clear offset between them. This similarity relies mainly on our new Almaden result, which is quite comparable with the pole established recently by *Hargraves et al.* [1987] on the Aïr ring complexes. It means that these two regions were moving in the same sense, at the same apparent velocity during this whole period. While we already suspected [*Perroud et al.*, 1984] that Spain was attached to Gondwana in the Early Ordovician, it seems that this conclusion can be extended to the whole Ordovician, Silurian and Early Devonian. On the other hand, the divergence of the curves at 380 Ma suggests that separation between Spain and Gondwana has occurred during the Devonian. However, it should be noted that this age of 380 Ma is not very precise, since it corresponds to the magnetization found in Siluro-Devonian Asturian redbeds [*Perroud*, 1984], whose age is controlled only by the positive fold test obtained with the Carboniferous deformation. Generally speaking, age control for pre-Devonian magnetizations is better, because both volcanics and redbeds results are available and concordant, than for younger magnetizations where rock type is mostly sedimentary.

The extremely high continental drift rates imposed by our results (up to 15-20 cm/yr) were already emphasized by *Bachtadse and Briden* [1990], who have obtained a maximum of 23 cm/yr in their compilation of Gondwana results. Such high velocities are not observed for the present-day plates, and it is not clear whether they are even possible. Several assumptions are necessary to estimate plate velocities from observed directions of magnetization: dipolar EMF, no polar wander, constant radius of Earth, correct age estimations, correct polarity assumptions, etc. Aberrant velocity estimations could therefore signify that one of them is not verified, or that the regime of the Paleozoic plate tectonic was very different from nowadays. In eitheir case, crucial questions for the understanding of the evolution of the Earth are opened, and the importance of such matters appeals for further investigations.

Acknowledgements. We thank our colleagues J. P. Cogné and M. Robardet for their assistance in the field, and the whole staff of the Almaden Polytechnic Mining School for their very friendly welcome. We thank too the direction of the Almaden Mines for their authorization to drill into an active gallery, and V. Courtillot who has accepted our request for using his laboratory in Paris. Journal of Geophysical Research editor S. L. Halgedahl and the two anonymous reviewers have provided very helpful comments on earlier versions of this paper.

REFERENCES

Bachtadse, V., and J. C. Briden, Palaeomagnetic constraints on the position of Gondwana during Ordovician to Devonian times, *Geol. Soc. Memoir, 12*, 43-48, 1990.

Bonhommet, N., P. R. Cobbold, H. Perroud, and A. Richardson, Paleomagnetism and cross-folding in a key area of the Asturian arc (Spain), *J. Geophys. Res., 86*, 1873-1887, 1981.

Briden, J. C., H. B. Turnell, and D. R. Watts, British palaeomagnetism, Iapetus ocean, and the Great Glen Fault, *Geology, 12*, 428-431, 1984.

Eichman, R. M., F.V. Saupé, and M. Schidlowski, Carbon and oxygen isotope studies in rocks of the vicinity of the Almaden mercury deposit (Province of Cuidad Real, Spain), in *Time and Srata Bound Ore Deposits*, edited by D. D. Klemm. and H. J. Schneider, 396-405, Springer-Verlag Berlin, 1977.

Ellwood, B. B., Anisotropy of magnetic susceptibility; empirical evaluation of instrumental precision, *Geophys. Res. Lett., 11*(7), 645-648, 1984.

Hargraves, R. B., E. M. Dawson, and F. B. Van Houten, Palaeomagnetism and age of mid-Palaeozoic ring complexes in Niger, West Africa, and tectonic implications, *Geophys. J. R. Astron. Soc., 90*, 705-729, 1987.

Leutwein, F., F. Saupé, J. Sonet, and E. Bouyx, Première mesure géochronologique en Sierra Morena. La granodiorite de Fontanosas (Province de Cuidad Real, Espagne), *Geol. Mijnbouw, 49*, 297-304, 1970.

Paris, F., and M. Robardet, Paléogéographie et relation ibéro-armoricaine au Paléozoïque anté-carbonifère, *Bull. Soc. Géol. France, 7*(19), 1121-1126, 1977.

Paris, F., and M. Robardet, Early Paleozoic palaeobiogeography of the Variscan regions, *Tectonophysics, 177*, 193-213, 1990.

Perroud, H., Contribution à l'étude paléomagnétique de l'arc Ibéro-Armoricain, *Bull. Soc. Géol. Bretagne, 14*, 1-114, 1982.

Perroud, H., Palaeomagnetism of Palaeozoic rocks from the Cabo de Penas, Asturia, Spain, *Geophys. J. R. Astron Soc., 75*, 201-215, 1983.

Perroud, H., A Devonian palaeomagnetic pole for Armorica, *Geophys. J. R. Astron. Soc., 77*, 839-845, 1984.

Perroud, H., Paléomagnétisme dans l'arc Ibéro-Armoricain et l'orogénèse Varisque en Europe Occidentale, thèse de doctorat d'état, Univ. de Rennes 1, Rennes, 1985.

Perroud, H., Paleomagnetic evidence for tectonic rotations in the Variscan mountain belt, *Tectonics, 5*, 205-214, 1986.

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Perroud, H., and N. Bonhommet, Palaeomagnetism of the Ibero-Armorican arc and the Hercynian orogeny in western Europe, *Nature, 292*, 445-448, 1981.

Perroud, H., R. Van der Voo, and N. Bonhommet, Paleozoic evolution of the Armorica plate on the basis of paleomagnetic data, *Geology, 12*, 579-582,1984.

Saupé, F., La série ordovicienne et silurienne d'Almaden (Province de Cuidad Real, Espagne). Point des connaissances actuelles, *Mem. BRGM, 73*, 355-365, 1971.

Saupé, F., G. Dunoyer de Segonzac, and M. Teichmuller, Etude du métamorphisme régional dans la zone d'Almaden (Province de Cuidad Real, Espagne) par la cristallinité de l'Illite et par le pouvoir reflecteur de la matière organique, *Sci. Terre, 21*, 251-269,1977.

Thellier, E, and O. Thellier, Sur l'intensité du champ magnétique terrestre dans le passé historique et géologique, *Ann. Geophys., 15*, 285-376,1959.

Tucholka, P., and J. Tessier, Interfacing the Digico flux-gate spinner magnetometer to an Apple IIe microcomputer, *Geophys. J. R. Astron. Soc., 89*, 705-712, 1987.

Van der Voo, R., The rotation of Spain: Paleomagnetic evidence from the Spanish Meseta, *Palaeogeogr. Palaeoclimatol. Palaeoecol., 3*, 393-416,1967.

Van der Voo, R., Paleomagnetic evidence for the rotation of the Iberian Peninsula, *Tectonophysics, 7*, 5-56, 1969.

Van der Voo, R., Paleozoic assembly of Pangea: A new plate tectonic model for the Taconic, Caledonian, Hercynian orogenies (abstract), *Eos Trans. AGU, 60*, 241, 1979.

Van der Voo, R., Paleozoic paleogeography of North America, Gondwana, and intervening displaced terranes: Comparison of paleomagnetism with paleoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull., 100*, 311-324, 1988.

Veitch, R. J., I. G.Hedley, and J. J. Wagner, Magnetic anisotropy delineator calibration error, *Geophys. J. R. Astron Soc., 75*, 407-409, 1983.

Verges J., Estudio del complejo volcano-sedimentario del Devonico y de la structura de la terminacion oriental del synclinal de Almaden (Cuidad Real), in *Libro Jubilar J.M. Rios*, vol. III, pp. 215-229, Madrid, 1983.

F. Calza, D. Kattach, and H. Perroud, Laboratoire de Géophysique Interne, CAESS (U.P.R. 4661 CNRS), Campus de Beaulieu, Université de Rennes 1, 35042, Rennes Cédex, France.

(Received March 2, 1990;
revised September 25, 1990;
accepted October 11, 1990.)

21562202b, 1991, B2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02226 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License